A comment on “Effects of 8-week endurance, strength, and coordination exercise interventions on attention in adolescents: a randomised controlled study”

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Original study: Altermann, W., & Gröpel, P. (2025). Effects of 8-week endurance, strength, and coordination exercise interventions on attention in adolescents: a randomised controlled study. *International Journal of Sport and Exercise Psychology*, 23(3), 487–500. DOI: 10.1080/1612197X.2024.2332280

Replication package: Figshare record named in the article (DOI 10.6084/m9.figshare.23703237; not accessed in this audit). The audited deposit holds two files: an SPSS data file with 120 rows and 22 variables (`pkg/Altermann_LongtermExercise&Attention.sav`) and the SPSS syntax for the reported analyses (`pkg/Study_Syntax.txt`).

Abstract

Altermann and Gröpel (2025) report a trial in which 96 adolescents completed eight weeks of endurance, strength or coordination training, or a reading control, and conclude that exercise raises d2 attention scores, most of all coordination training. The deposit holds one SPSS data file with 120 rows and the SPSS syntax. We recomputed Tables 1 and 2 and every Results statistic in R. Of 150 printed values, 145 agree at printed precision, three differ by one unit in the last digit, and two do not: the control group’s pretest means for concentration (157.40 printed, 157.00 in the data) and speed (174.52, 174.42), which no 24 integer scores can produce. The exercise-versus-control result survives every alternative analysis we ran. The claim that coordination works best does not. The coordination contrast ($p = .018$) exceeds the article’s own Bonferroni threshold of .0125, coordination is not distinguishable from endurance ($p = .32$), and the contrast’s status depends on the specification and on three records whose scores contradict their own error counts. Smaller findings concern identical physical-education-hour splits across groups, the randomised N in the Abstract, unreported homogeneity tests, deposit documentation and quasi-identifiers in the data file. One published conclusion is narrowed: that coordination training works best. The exercise-versus-control effect is not.

Section 1. Introduction

Altermann and Gröpel (2025) randomised students at one Viennese secondary school to eight weeks of high-intensity interval running (labelled endurance), a strength circuit, a coordination circuit or a reading control. Sessions lasted 50 minutes and ran twice a week. The d2-R attention test was taken one week before and one week after the programme. The authors analyse concentration, error and speed scores in 4 (group) by 2 (phase) mixed ANOVAs with planned contrasts. They conclude that all three exercise groups gained more than the control group, that the coordination group gained most and the strength group least, and that physical education teachers should add coordinative tasks.

This comment was produced under the Institute for Replication's AI-assisted reproducibility protocol (Brodeur et al., 2026; Institute for Replication, 2026). The deposit has one SPSS data file and one SPSS syntax file. SPSS was not available, so the syntax was not executed. Every printed statistic was instead recomputed from the data file in R, with the within-subject contrasts written as the change-score contrasts that the syntax's LMATRIX and MMATRIX lines specify. We also tested scores against their defining identities, searched for duplicated and out-of-range values, tested printed means for arithmetic attainability, compared group composition with the Methods and Figure 1, and re-estimated the contrasts under baseline adjustment, unequal variances and permutation. Quasi-identifiers were counted without printing any row. Not verified: the figshare record and its licence, the AsPredicted preregistration (no network access), and anything that needs materials absent from the deposit, namely d2 answer sheets, the allocation spreadsheet and attendance records. Each finding has a check in `verification_snippets/` and its scripts are in `evidence/`.

Section 2. What reproduces: 145 of 150 printed values, the flow chart, and the sample-size statement

The table below compares every printed value in the article's Tables 1 and 2 and its Results text with the deposit. Deposit values are rounded half up to the printed precision.

Table 1. Printed values in the article's Tables 1 and 2 and Results text compared with values computed from the deposited data file, by block. Deposit values are rounded half up to the printed precision.

Block of printed values	Compared	Agree	One unit off	Disagree
Table 1: leisure hours, PE hours and grades (means, SDs, F, p, η^2)	33	33	0	0
Table 1: perceived exertion	8	8	0	0

Block of printed values	Compared	Agree	One unit off	Disagree
Table 2: group means	24	22	0	2
Table 2: group SDs	24	24	0	0
Table 2: F, p and η² columns	27	26	1	0
Results text: F values	16	14	2	0
Results text: p values	16	16	0	0
Results text: exertion pairwise p values	2	2	0	0
All blocks	150	145	3	2

The three one-unit differences are the F for the endurance contrast on concentration (0.28 printed, 0.2748 computed), the F for the coordination contrast on the error score (8.80, 8.795) and the group η² for the error score (.04, .035). Each printed value equals the computed one rounded to three decimals and then again to two, which fits transcription of SPSS's three-decimal output. The two disagreements are in Section 5. The degrees of freedom match the analysed n (92 is 96 minus 4 groups; 69 is 72 minus 3 groups). The pairwise exertion p values are .0183 (printed as "p < .018"), .0003 (printed "p < .001") and .5625 (printed .563).

Figure 1's counts equal the deposit's: 34, 31, 28 and 27 randomised, 10, 7, 4 and 3 lost, 24 analysed per group (read from the rendered figure). The sample-size statement reproduces. With f = .25, α = .05, power .95, four groups, two measurements and a correlation of .5 (the G*Power default, which the article does not state), a noncentral-F calculation requires 73 participants, and 76 is the next multiple of four groups. The descriptive statements also hold: 48 women and 48 men, six participants of each age in every group, 80.2% active outside school, 24 of 120 lost (20%), and training blocks that add to 25 minutes within 50-minute sessions and 16 sessions in total.

The package has specific strengths. It includes the lost participants' pretest scores, which makes Figure 1 checkable. The syntax states each contrast and its transformation matrix, so the tests can be rewritten outside SPSS. The article prints degrees of freedom, effect sizes and contrast weights.

Section 3. The coordination advantage is not established by the article's own test

What. The Methods set a Bonferroni-adjusted threshold of .0125 for the contrast tests. The Results report the coordination contrast on concentration as F(1, 92) = 5.84, p = .018, and call it marginally significant. The Abstract, Discussion and Conclusion then say coordination training had the greatest effect, "rather than" endurance, and advise teachers to add coordinative tasks. The article never compares coordination with endurance: its endurance contrast sets endurance against the average of the other two exercise groups. In the deposit, mean concentration gains are 47.1 points (coordination), 41.1 (endurance), 30.0 (strength) and 19.0 (control). Coordination minus endurance is 6.05 points, p = .32. Coordination minus strength is 17.1 points, p = .006. The contrast that carries the coordination result is therefore driven mostly by the low strength group.

Table 2. Coordination contrast on concentration performance (article threshold .0125) and coordination-versus-endurance comparison, by specification and by treatment of three inconsistent records. Two-sided p values.

Treatment of three records	Contrast p: pooled (article)	Contrast p: unequal variances	Contrast p: baseline-adjusted	Contrast p: permutation	Coordination vs endurance p: pooled	Coordination vs endurance p: baseline-adjusted
As published	.0177	.0392	.0226	.033	.32	.09
Scores recomputed from counts	.0102	.0270	.0139	.021	.25	.07
Three rows dropped	.0115	.0306	.0215	.026	.25	.11

Evidence. The article's contrast reproduces (p = .0177). It stays above .0125 with unequal variances (.039; the change-score SD runs from 12.3 in the control group to 22.5 in the coordination group), with baseline adjustment (.023) and under 20,000 permutations (.033). Within groups the pretest correlates −0.72 with the change score, the pretest-posttest correlation runs from .23 (coordination) to .85 (control), and the strength group started 12.5 points above the endurance group, so change-score contrasts partly reflect baseline differences. The contrast falls below .0125 only in the pooled-variance analysis after the three records of Section 4 are recomputed or dropped. Coordination against endurance stays above p = .06 under every specification and treatment (smallest .068). The strength contrast (p = .0042) becomes .0615 with baseline adjustment. The three exercise groups differ in gain (F(2, 69) = 4.11, p = .0205), mainly through the strength group. For the error score the coordination contrast is p = .0038 as published, .018 with unequal variances and .10 with baseline adjustment. Scripts: `evidence/02_coordination_claim.R` and `evidence/04_variants.R`.

Impact. The exercise-versus-control contrast stays below .001 under all four specifications on the published data, and under the pooled test after either treatment of the three records, so the first hypothesis stands. The statement that coordination has the greatest effect, and the advice built on it, narrows to this: coordination has the numerically largest gain, a larger gain than strength, and a gain not distinguishable from endurance. The Results text calls the contrast marginal. The Abstract, Discussion and Conclusion do not.

Remedy. Report coordination against endurance and against strength directly, with baseline-adjusted estimates and unequal-variance errors, and present the ordering of exercise types as exploratory.

Section 4. Three records carry scores that contradict their own error counts

What. In the data file the concentration score equals speed minus the omission and commission counts, and the error percentage equals those counts divided by speed. This holds in 119 of 120 pretest rows and 94 of 96 posttest rows. One pretest row (coordination group) and two posttest rows (one endurance, one control) break both identities, and all three are in the analysed sample. In the two posttest rows the concentration score is not an integer, although it is a count minus a count, and both rows imply one and the same non-integer error count. In all three rows the stored omission and commission counts exceed the error count that the scores imply, by 10.85 to 19.85 errors.

Evidence. `evidence/03_integrity_checks.R` and `evidence/04_variants.R`. Rounding cannot produce a non-integer count. Without the answer sheets we cannot tell which stored values are right, so the analysis is repeated twice: with scores recomputed from the counts in every row, and with the three rows dropped. The phase-by-group F for concentration moves from 10.07 to 11.22 (10.94 with the rows dropped) and for the error score from 8.06 to 8.23 (7.47). The control group’s posttest concentration mean moves from 175.99 to 175.17. The interaction p stays below .001 in every case.

Impact. Several Table 2 means and F values shift slightly. No conclusion changes by itself. The coordination contrast moves as described in Section 3.

Remedy. Check the three records against the d2 answer sheets, correct the data file and rerun the syntax.

Section 5. Two control-group pretest means in Table 2 cannot arise from 24 integer scores

What. Table 2 prints 157.40 for the control group’s pretest concentration mean and 174.52 for its pretest speed mean. Both scores are integers and the group has 24 participants, so a mean must be a multiple of 1/24. The nearest attainable values are 157.38 and 157.42, and 174.50 and 174.54. The deposit gives 157.00 and 174.42, and the printed SDs (23.26, 20.81) match. The

other 10 integer-valued means in Table 2 are attainable (Brown and Heathers, 2017 describe the test).

Evidence. `evidence/01_reproduce_tables.R` and `evidence/03_integrity_checks.R`.

Impact. None on any test statistic, which reproduces from the data. The control gain in concentration is 19.0 points, where the printed means imply 18.6.

Remedy. Replace 157.40 with 157.00 and 174.52 with 174.42 in Table 2.

Section 6. The analysed groups share one weekly PE-hour split, and the Table 1 note misstates one group

What. Table 1 gives weekly physical-education hours as 3.25 (1.33) in all four groups, F = 0.00, p = 1.00. In the data every analysed group has the same split: 12 participants with 2 hours, 6 with 4 hours and 6 with 5 hours. The Methods stratify on age and gender only, and those strata do not fix PE hours: all eight age-by-gender strata contain at least two values. The probability that four random groups of 24 share one split is 0.0002 (exact, complete randomisation) and 0.0005 within the age-by-gender strata (one million shuffles). Among all 120 randomised participants the splits differ (18/7/9, 16/8/7, 14/7/7 and 14/7/6). Separately, the Table 1 note says each group has three women and three men at each age. In the coordination group the age-15 stratum has 2 men and 4 women and the age-16 stratum has 4 men and 2 women. The other 28 of 32 group-by-age-by-gender cells hold exactly three.

Evidence. `evidence/03_integrity_checks.R`.

Impact. None on the tests. The Table 1 test of PE hours cannot detect imbalance in a variable that is identical across groups. The article says further recruits replaced the 24 lost participants until each group had 24 with equal age and gender distribution. It does not mention the equality on PE hours, and the deposit does not show how the analysed participants were chosen. We draw no inference about how the equality arose.

Remedy. Correct the Table 1 note, state how the analysed participants were selected, and say whether PE hours entered the allocation.

Section 7. The Abstract's N = 96 is the analysed sample; 120 were randomised

What. The Abstract says adolescent students (N = 96) were randomised. Figure 1 and the deposit show 120 randomised (34 endurance, 31 strength, 28 coordination, 27 control) and 24 lost to follow-up (10, 7, 4 and 3), leaving 24 analysed per group. The loss rates are 29.4%, 22.6%, 14.3% and 11.1%. The 24 lost participants have no posttest scores, so every analysis in the article, and every analysis possible from the deposit, compares completers. Loss rates do not differ significantly (Fisher p = .30). The lost participants' pretest error scores are lower than the

completers' (8.6 against 11.3, Welch p = .022). The Discussion's limitations do not mention attrition.

Evidence. `evidence/03_integrity_checks.R` and `evidence/05_baseline_attrition_levene.R`.

Impact. None on the tests. The estimates describe participants who completed all 16 sessions and the posttest, not all randomised participants.

Remedy. State 120 randomised and 96 analysed in the Abstract, give the per-group losses in the text, and add attrition to the limitations.

Section 8. Minor issues

Homogeneity tests and baseline differences are computed but not reported. The syntax requests Levene tests in five analyses and Bonferroni post-hoc comparisons in four. The article reports neither. Levene tests on the posttest scores reject equal variances for concentration (p = .012) and for the error score (p < .001). The largest posttest SD is about twice the smallest in both. The four analysed groups also differ in pretest error score, F(3, 92) = 2.93, p = .038 (means 11.54, 9.65, 13.85 and 10.14), a baseline difference the article does not report. Evidence: `evidence/05_baseline_attrition_levene.R`. Impact: none on published values; both results bear on Section 3. Remedy: report the tests and a baseline-adjusted analysis.

The deposit has no documentation and its syntax opens no data file. The deposit holds two files, with no README, codebook or licence. The syntax has no command that opens the data file, so it runs only on a dataset opened by hand. The data file name contains an ampersand. The file header records SPSS 27 on macOS, while the article states SPSS 26.0, and we could not test whether the syntax runs under 26. The strength group's value label and two variable names carry spelling variants. Evidence: `evidence/03_integrity_checks.R`. Impact: none. Remedy: add a GET FILE line, a README and a codebook, and state the licence. The GET FILE line was not run because SPSS was unavailable.

The data file carries quasi-identifiers, and the article states neither anonymisation nor consent for open release. `pkg/Altermann_LongtermExercise&Attention.sav` holds 120 rows and 22 columns for participants aged 15 to 18. The columns are a sequential row number, gender, age in years, weekly PE hours, a leisure-sport indicator and hours, the German, mathematics and English grades with their mean, and the d2 scores. There is no name, e-mail address, date, address or free text, and the file contains no e-mail-like string. No row is unique on age and gender, one row is unique with PE hours added, and 107 of 120 rows are unique with the three subject grades added (69 with the mean grade in their place). The article says participants and their parents consented but not that consent covered open sharing, and it does not claim anonymisation. The exposure is indirect and would need information from outside the file, so the urgency is routine. Evidence: `evidence/03_integrity_checks.R`. Impact: none. Remedy: drop the German, Math and English columns, which the syntax does not use and Table 1's grade row does not need (that row reproduces from the mean grade, F = 1.57, p = .201), and replace the file in the repository. After that change 69 rows remain unique on the remaining combination. Rounding the mean grade to a whole grade would lower this to 21 but would change the Table 1 grade statistics. Repository procedures were not checked.

Section 9. Hypotheses tested and not confirmed

- **Copied or duplicated rows.** No row repeats another once the row number is excluded. One pair of rows shares all five pretest scores. About 0.8 pairs would share the speed, omission and commission counts by chance if those counts were independent. No block of rows repeats at any offset.
- **Out-of-range values.** Age, grades and perceived exertion lie within their scales, and the mean grade equals the mean of the three subject grades in all 120 rows.
- **Signatures of random data.** Terminal digits of the pretest concentration and speed scores do not depart from uniformity ($p = .11$ and .55). The expected structure is present: concentration and speed correlate .91, the error percentage and omissions .92, and pretest and posttest concentration .44.
- **Sample size misstated.** The 76 reproduces (Section 2).
- **Differential attrition.** Loss rates by group differ in the data (29% to 11%) but not significantly (Fisher $p = .30$).
- **Dependence on software defaults.** The R re-implementation reproduces SPSS's Type III results and contrasts at printed precision, so no value depends on an estimator default. The SPSS version was not tested.
- **Fragility of the main result.** The exercise-versus-control contrast on concentration stays below .001 under pooled, unequal-variance, baseline-adjusted and permutation tests on the published data, and under the pooled test after either treatment of the three records. All four groups improve (paired $p < .000001$ each), including the control group.
- **PE hours fixed by age and gender.** Rejected: all eight strata contain at least two PE values.
- **Live repository and preregistration.** Not checked; there was no network access.

Section 10. Conclusion

The deposit reproduces the article's analyses at printed precision, with two exceptions: two control-group baseline means in Table 2 that the data do not support, and three values that differ by one unit in the last digit. The main result, that exercise groups gained more concentration than the control group, holds under baseline adjustment, unequal variances and permutation, and under the pooled test after either correction of three inconsistent records.

The ranking of exercise types holds less well. The article does not test coordination against endurance, and the deposit shows no significant difference. The coordination contrast clears the article's own threshold in two of twelve specification-and-treatment combinations. This comment does not establish which values in the three inconsistent records are correct, how the 96 analysed participants were selected, or whether the figshare record matches the deposit.

Data and Code Availability

The audited deposit consists of `pkg/Altermann_LongtermExercise&Attention.sav` and `pkg/Study_Syntax.txt`. The article names a figshare record (DOI 10.6084/m9.figshare.23703237) and an AsPredicted preregistration. Neither was accessed. The deposit states no licence, and the licence of the figshare record was not checked. The scripts that reproduce every number in this comment are in `evidence/`. The checks a replicator can run against the deposit are in `verification_snippets/`. Both read the deposit without modifying it.

References

Altermann, W., & Gröpel, P. (2025). Effects of 8-week endurance, strength, and coordination exercise interventions on attention in adolescents: a randomised controlled study. *International Journal of Sport and Exercise Psychology*, 23(3), 487–500. https://doi.org/10.1080/1612197X.2024.2332280

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Institute for Replication (2026). Reproducing with AI Across the Expertise Ladder. Pre-analysis plan, OSF: https://osf.io/dkfzt/ · https://github.com/I4Replication/AI-vertical

Appendix A. Summary of findings

Table 3. Summary of findings.

#	Finding	Status	Changes a published conclusion?	Section
1	The claim that coordination training has the greatest effect rests on a contrast that fails the article's own threshold and on no direct comparison with endurance.	CONFIRMED	Yes	Section 3
2	Three records have concentration and error scores that contradict their own omission and commission counts, two with non-integer concentration scores.	CONFIRMED	No	Section 4
3	Two control-group pretest means in Table 2 (concentration 157.40, speed 174.52) cannot arise from 24 integer scores; the data give 157.00 and 174.42.	CONFIRMED	No	Section 5
4	The analysed groups have identical weekly PE-hour splits, which the described stratification does not explain, and the Table 1 note misstates the coordination group's strata.	CONFIRMED	No	Section 6
5	The Abstract's N = 96 is the analysed sample; 120 were randomised, 24 were lost (29%, 23%, 14%, 11% by group), and the analysis uses completers only.	CONFIRMED	No	Section 7
6	Homogeneity tests and baseline group differences computed in the syntax are not reported; posttest variances differ and pretest error scores differ across groups.	CONFIRMED	No	Section 8
7	The deposit has no README, codebook or licence, its syntax opens no data file, and the data file records a different SPSS version from the article.	CONFIRMED	No	Section 8
8	The data file holds individual school grades, age, gender and PE hours that make 107 of 120 rows unique; no anonymisation or consent for open	CONFIRMED	No	Section 8

#	Finding	Status	Changes a published conclusion?	Section
	release is stated.			